SCREAM DETECTION AND ALERT SYSTEM

Project Overview:
The Scream Detection System detects distress or ambiguous screams in real-time and alerts emergency contacts, and provides location details. It aims to enhance security in emergency situations and provide real time assistance in handling emergencies.

Product Scope:
The system is an ML model powered tool that:
-Captures live audio.
-Detects ambiguous screams using a trained deep learning algorithm.
-Upon detection the system sends an alert message via sms.

Product Functions

1. **Real-time audio capture** – Records live sound using a microphone.
2. **Feature extraction** – Extracts Mel-Frequency Cepstral Coefficients (MFCC) from the audio.
3. **Scream classification** – Uses a TensorFlow-based deep learning model to classify the sound.
4. **Location retrieval** – Obtains the user's real-time location using Google Geolocation API.
5. **Alert generation** – Sends an SMS alert to a predefined contact via Twilio.

Product Constraints

- Requires a working microphone for real-time detection.
- Internet connectivity is mandatory for API calls.
- Model accuracy limitations (may require fine-tuning).

System Components

- **Audio Input** – Captures sound using sounddevice.
- **Feature Extraction** – Uses librosa to extract MFCCs.
- **ML Model** – A pre-trained TensorFlow .h5 model for scream classification.
- **API Calls** – Google Geolocation API for location and Twilio for SMS.
- **User Interface** – Command-line or GUI for monitoring alerts.

Technologies Used:

Programming Language : Python

Audio Processing : Sounddevice, librosa, numpy

ML libraries : TensorFlow

Cloud Service : Google Geolocation API

Alert System : Twillo SMS API

Execution Plan

Phase 1: Data Preparation & Model Training

- Collect scream and non-scream audio samples.
- Extract MFCC features using librosa.
- Train a neural network in TensorFlow/Keras.

Phase 2: Real-time Audio Processing

- Capture live audio using a microphone.
- Extract MFCC features in real-time.
- Pass extracted features to the trained model.
- Classify the audio as "Scream" or "No Scream".

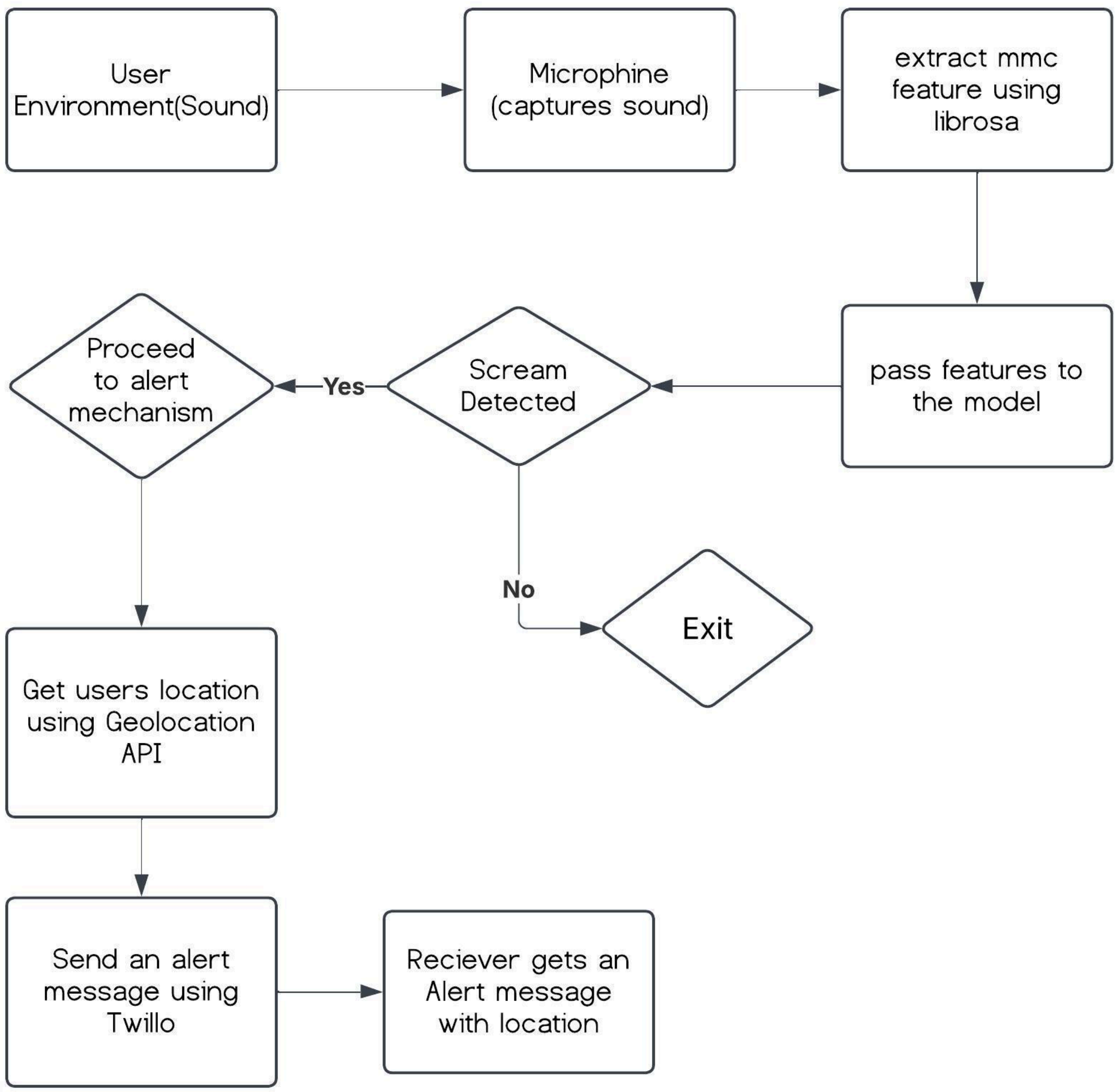
Phase 3: Alert Mechanism

- If a scream is detected:
    - Fetch the user's location via Google Geolocation API.
    - Send an SMS alert using Twilio with the location details.

Phase 4: Deployment & Testing

- Test the system with live audio.
- Fine-tune the model if necessary.
- Integrate the ml model to mobile application using kiwi and android app development by accessing mobile's microphone and device location.

Data Flow Diagram



Appendix

- **Technologies Used**: Python, TensorFlow, Librosa, Twilio API, Google API.
- **Security Measures**: API key protection, encrypted communication.
- **Future Enhancements**:
    - Mobile app integration.
    - Background noise filtering.
    - Smartwatch compatibility.